

(1001CJA100120010)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

04-10-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	B,C,D	B,C,D	A,B,C	B,C	B,C	A,C,D	D	A	C
	Q.	11	12	13							
	A.	A	B	C							
SECTION-II	Q.	1	2	3	4	5					
	A.	3.00	4.00	3.00	50.00	800.00					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	C,D	A,B,C,D	A,B,D	A,D	A,B,D	A,B,C,D	C	C	D
	Q.	11	12	13							
	A.	B	D	A							
SECTION-II	Q.	1	2	3	4	5					
	A.	2.00	7.00	5.00	4.00	13.15					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,D	A,B,C	A,C,D	A,B,C,D	A,B,C	A,B,C	C	B	D
	Q.	11	12	13							
	A.	D	C	D							
SECTION-II	Q.	1	2	3	4	5					
	A.	0.75	0.20	0.25	20.25	0.50					

(1001CJA100120011)

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**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

UNIT TEST

04-10-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	C	C	D	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	C	B	B	A	A	A	B	C
SECTION-II	Q.	1	2	3	4	5					
	A.	10.00	8.00	5.00	5.00	15.00					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	A	B	C	B	C	D	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	B	B	D	C	B	D	C	B
SECTION-II	Q.	1	2	3	4	5					
	A.	4.00	4.00	5.00	4.00	17.69 to 17.71					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	D	B	D	C	B	D	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	A	C	B	D	C	C	C	D
SECTION-II	Q.	1	2	3	4	5					
	A.	1.50	20.00	1.56 or 1.57	3.00	0.00					

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A, C)**

$$\text{Sol. } mgy = \frac{1}{2}ky^2 \Rightarrow y = \frac{2mg}{k}$$

$$W_{N.C} + W_{mg} + W_{spring} = 0$$

$$W_{NC} = -mg(y_{eq}) + \frac{1}{2}ky_{eq}^2$$

$$y_{eq} = \frac{mg}{k}$$

$$W_{NC} = -\frac{(mg)^2}{2k}$$

2. **Ans. (B,C,D)**

$$\text{Sol. (B) Mass} \propto \frac{1}{(\text{radius})^2}$$

$$\Rightarrow \frac{m}{M_C} = \left(\frac{R}{R/2}\right)^2$$

$$\Rightarrow m_C = \frac{m}{4}$$

Mass of remaining part

$$= m - \frac{m}{4} = \frac{3m}{4}$$

$$(C) y = \frac{m \times 0 - \frac{m}{4} \times \frac{R}{2}}{m - \frac{m}{4}}$$

$$y = -\frac{R}{6}$$

(D) Object is symmetric about x-axis

3. **Ans. (B, C, D)****Sol.** Linear momentum & mechanical energy conserved during collision.

Only conservative forces act during the collision due to which mechanical energy is conserved.

4. **Ans. (A, B, C)****Sol.** (A) Net external force on the given system is zero, so momentum of the system is conserved in ground frame.

$$(B) \vec{P}_{\text{system}} = (m_1 + m_2) \cdot \vec{v}_{\text{com}}$$

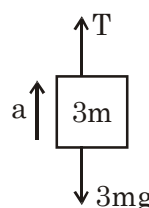
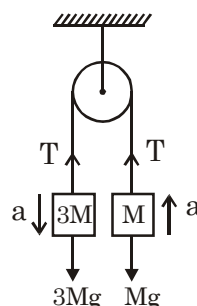
In COM frame $\vec{v}_{\text{com}}$ is equal to zero. $\Rightarrow \vec{P}_{\text{system}}$ in COM frame is always zero.

$$(C) \vec{F}_{\text{Ext}} = 0 \Rightarrow \vec{a}_{\text{Com}} = 0 \Rightarrow \vec{v}_{\text{Com}} \text{ is constant.}$$

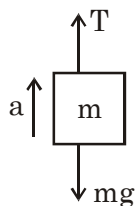
$$KE_{\text{Com}} = \frac{1}{2}(m_1 + m_2) \cdot v_{\text{Com}}^2 \text{ is remains constant.}$$

$$(D) \vec{K} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (\vec{v}_{12})^2$$

KE in COM frame changes into spring potential energy

5. **Ans. (B,C)****Sol.**

$$\Rightarrow 3mg - T = 3ma \dots(i)$$



$$\Rightarrow T - mg = ma \quad \dots(ii)$$

Add eq.(i) + eq.(ii)

$$2mg = 4ma$$

$$\Rightarrow a = g/2$$

From eq.(ii)

$$T = mg + mg/2 = 3mg/2$$

$$\vec{a}_{\text{com}} = \left[\frac{Ma + 3M(-a)}{3M + M} \right] \hat{j} = \frac{-2Ma}{4M} \hat{j} = \frac{-g}{4} \hat{j}$$

$$\vec{F}_{\text{Ext}} = [4M] \vec{a}_{\text{Com}}$$

$$\Rightarrow \vec{F}_{\text{Ext}} \neq 0$$

6. Ans. (B,C)

Sol. Since external force on the system (A + B + plank) in horizontal direction is zero therefore centre of mass of the system remain at rest.

Let displacement of plank is x, then

$$60(4 - x) - 50(4 + x) = 90^\circ x$$

$$40 = 200x$$

$$x = 0.2 \text{ m}$$

7. Ans. (A,C,D)

Sol. For equilibrium

$$F = 0, \frac{-dU}{dx} = 0$$

(A) At point 3

$$F = 0 \text{ and } \frac{dF}{dx} < 0$$

$\Rightarrow$ stable

At point b

$$\frac{dU}{dx} = 0, \frac{d^2U}{dx^2} > 0$$

$\Rightarrow$ stable

(B) At point 2

$F \neq 0$, So there is no equilibrium.

(C) $P = \vec{F} \cdot \vec{V}$

$F = 0$ At point 1, 3 and b

Power delivered by force = zero.

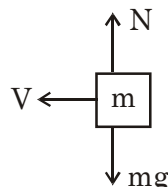
(D) Potential energy is minimum at point b, so KE is maximum.

8. Ans. (D)

Sol. At point C

$$W_g = \Delta KE = \frac{1}{2}mv_c^2 - 0$$

$$v_c^2 = 10Rg \Rightarrow v_c = \sqrt{10Rg}$$



$$\Rightarrow N - mg = \frac{m}{R}[10Rg]$$

$$N = 11mg$$

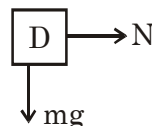
Tangential acceleration (a_t) is zero at point C

9. Ans. (A)

Sol. At point D

$$mg(4R) = \frac{1}{2}mv_D^2 - 0$$

$$v_D = \sqrt{8Rg}$$



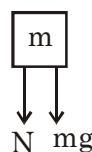
$$\Rightarrow N_D = \frac{mv_D^2}{R} = 8mg$$

$$\Rightarrow mg = ma_t$$

$$a_t = g$$

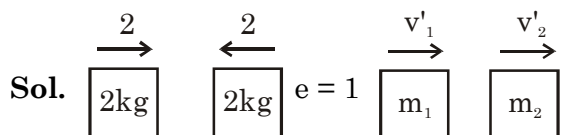
10. Ans. (C)

$$\text{Sol. } mg(3R) = \frac{1}{2}mv_E^2 \Rightarrow v_E = \sqrt{6gR}$$



$$\Rightarrow N + mg = \frac{m(6Rg)}{R} \Rightarrow N = 5mg$$

11. Ans. (A)



$$\vec{v}_{\text{Com}} = \frac{2 \times 2 + 2 \times (-2)}{4} = 0$$

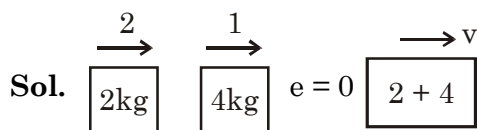
$$v'_2 = 2v_{\text{Com}} - v_2 = 2 \times 2 - (-2) = +2 \text{ m/s}$$

$$KE_{\text{Loss}} = 0 \text{ [For elastic collision]}$$

$$PE_{\text{max}} = [KE]_i - [KE]_{\text{Com}}$$

$$PE_{\text{max}} = 8 \text{ J}$$

12. Ans. (B)



$$2 \times 2 + 4 \times 1 = 6 \times v$$

$$\Rightarrow v = \frac{8}{6} = \frac{4}{3} \text{ m/s}$$

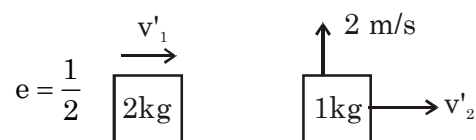
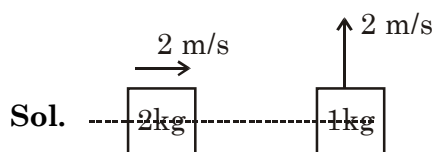
$$KE_i = \frac{1}{2} \times 2(2)^2 + \frac{1}{2} \times 4 \times (1)^2 = 6 \text{ J}$$

$$KE_F = \frac{1}{2} \times 6 \times \left[\frac{4}{3}\right]^2 = 3 \times \frac{16}{9} = \frac{16}{3} \text{ J}$$

$$KE_{\text{Loss}} = KE_i - KE_F = 6 - \frac{16}{3} = \frac{2}{3} \text{ J}$$

$$PE_{\text{max}} = KE_i - KE_{\text{COM}} = \frac{2}{3} \text{ J}$$

13. Ans. (C)



v_{com} along line of impact

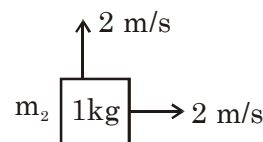
$$v_{\text{com}} = \frac{2 \times 2 + 1 \times 0}{3} = \frac{4}{3} \text{ m/s}$$

$$v'_1 = (1 + e)v_{\text{com}} - ev_1$$

$$= \frac{3}{2} \times \frac{4}{3} - \frac{1}{2} \times 2 = 1 \text{ m/s}$$

$$v'_2 = (1 + e)v_{\text{com}} - ev_2$$

$$= \frac{3}{2} \times \frac{4}{3} = 2 \text{ m/s}$$



$$\sqrt{(2)^2 + (2)^2} = 2\sqrt{2} \text{ m/s}$$

$$KE_i = \frac{1}{2} \times 2(2)^2 + \frac{1}{2} \times 1(2)^2 = 6 \text{ J}$$

$$KE_F = \frac{1}{2} \times 2(1)^2 + \frac{1}{2} \times 1 \times (2\sqrt{2})^2 = 5 \text{ J}$$

$$\Delta KE_{\text{Loss}} = KE_i - KE_F = 6 \text{ J} - 5 \text{ J} = 1 \text{ J}$$

$$PE_{\text{max}} = KE_i - KE_{\text{Com}}$$

$$= \frac{1}{2} \times 2 \times (2)^2 - \frac{1}{2} \times 3 \times \left[\frac{4}{3}\right]^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3} \text{ J}$$

SECTION-II

1. Ans. 3.00

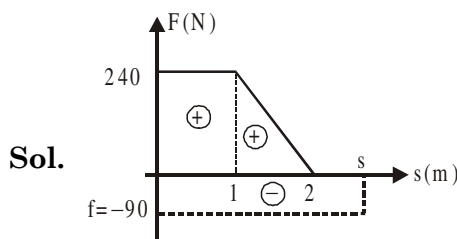
Sol. $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \Rightarrow v_1 + 3v_2 = 4 \dots(i)$

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow v_2 - v_1 = 4 \dots(ii)$$

By solving above equations $v_2 = 2 \text{ m/s}$

Momentum of B = $(1.5)(2) = 3 \text{ kg m/s}$

2. Ans. 4.00



The block will come to rest when total area

$$\text{will become zero } \frac{1}{2}(20)(1+2) - 90(s) = 0$$

$$\Rightarrow s = 4 \text{ m}$$

3. Ans. 3.00

Sol. Velocity of B after collision will be

$$V = \frac{3}{2} \frac{2mv_0}{3m} = v_0$$

Tension just after the collision is

$$T = \frac{mv_0^2}{\ell} + mg = 40$$

$$\Rightarrow v_0^2 = 30 \times 0.3$$

$$\Rightarrow v_0 = 3 \text{ m/s}$$

4. Ans. 50.00

Sol. $\frac{1}{2}mu_1^2 + \frac{1}{2}mu_2^2 = \frac{1}{2}kx^2 + \frac{1}{2}(2m)v^2; v = \frac{v_1 + v_2}{2}$

$$\Rightarrow x = \frac{1}{2}m = 50 \text{ cm}$$

5. Ans. 800.00

Sol. By conservation of momentum

$$-200 \times v_1 + 2 \times v_2 = 0$$

$$100 v_1 = v_2 \quad \dots (1)$$

$$v_1 = v_2 / 100$$

$$\left(\frac{v_2}{100} \right) \cdot t = 8$$

$$t = \frac{800}{v_2}$$

$$v_2 \times t = x$$

$$x = v_2 \times \frac{800}{v_2}$$

$$x = 800 \text{ m}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A, B, C, D)

2. Ans.(C,D)



50 ml, 1M 10 ml, 2M

$$1 \times 50 \times \text{n-factor} = 2 \times 10 \times 5$$

$$\text{n-factor} (\text{Sn}^{+2}) = 2$$

So Sn^{+2} is oxidized to Sn^{+4}

3. Ans. (A,B,C,D)

Sol.

$$V = \frac{nRT}{P} : V_{\text{H}_2} = \frac{\left(\frac{1}{2}\right) \times 0.0821 \times (273 + 273)}{1} = 22.4 \ell$$

$$V_{\text{He}} = \frac{\left(\frac{2}{4}\right) \times 0.0821 \times (273 + 273)}{1} = 22.4 \ell$$

$$V_{\text{CH}_4} = \frac{\left(\frac{3}{16}\right) \times 0.0821 \times (273 + 273)}{1} = 8.4 \ell$$

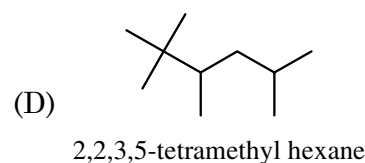
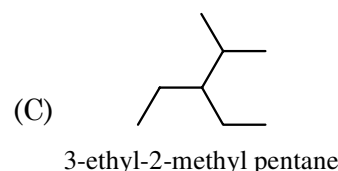
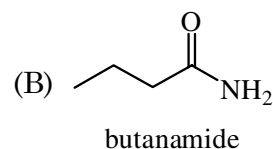
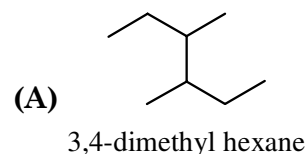
$$V_{\text{SO}_2} = \frac{\left(\frac{4}{64}\right) \times 0.0821 \times (273 + 273)}{1} = 2.8 \ell$$

4. Ans (A, B, D)

5. Ans. (A,D)

6. Ans (A, B, D)

7. Ans. (A, B, C, D)



8. Ans.(C)

9. Ans.(C)

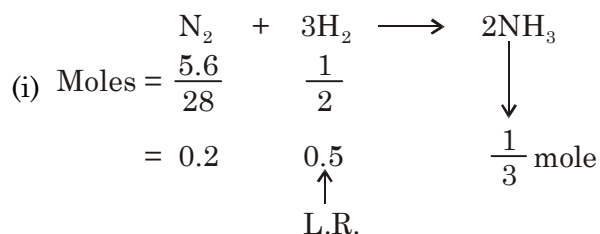
10. Ans.(D)

11. Ans.(B)

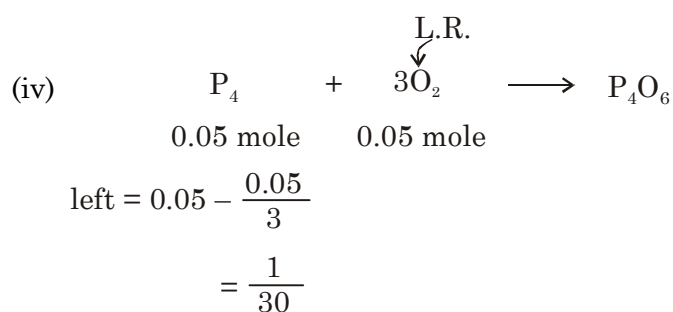
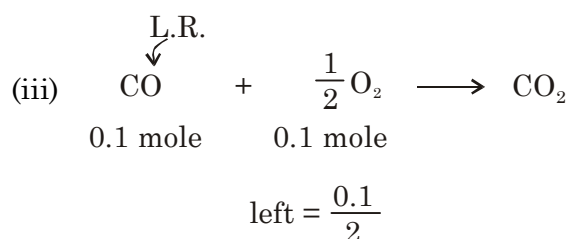
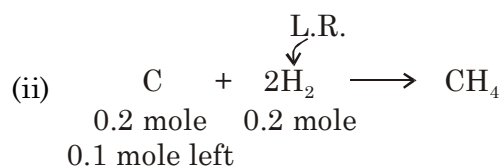
12. Ans.(D)

13. Ans. (A)

Sol.



$$\text{N}_2 \text{ left} = 0.2 - \frac{0.5}{3} = \frac{1}{30}$$


SECTION-II

1. Ans.(2.00)

$$\text{Sol. } P \times V = \frac{4}{M} \times R \times T \rightarrow (1)$$

$$P \times V = \frac{3.2}{M} \times R \times (T+50) \rightarrow (2)$$

$$\frac{(1)}{(2)} \Rightarrow 1 = \frac{4T}{3.2(T+50)}$$

$$T = 200 \text{ K} = 2.00 \times 10^2 \text{ K}$$

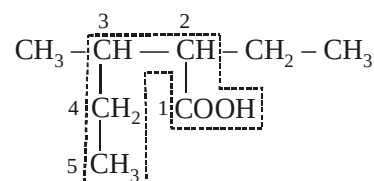
2. Ans (7.00)

number of Ring = 5

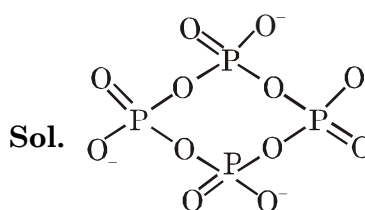
number of II bond = 2

$$\therefore \text{DBE} = 7$$

3. Ans. (5.00)



4. Ans.(4.00)



5. Ans. (13.15)

$$\text{Sol. } x = 4; y = 21; z = 1.30$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (B,C,D)**

Sol. System has unique solution

$$\Rightarrow \begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & \lambda \\ 1 & \lambda & 1 \end{vmatrix} \neq 0 \Rightarrow \lambda \neq -\frac{1}{2}, 1$$

2. **Ans. (A,D)**

Sol. $\frac{a}{3} = \frac{b}{4} = \frac{c}{5} = \lambda$

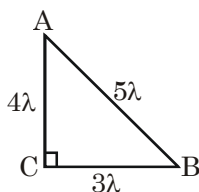
$$a = 3\lambda, b = 4\lambda, c = 5\lambda,$$

$$\lambda > 0$$

$$\begin{aligned} \text{Hence } a^2 + b^2 &= 9\lambda^2 + 16\lambda^2 \\ &= 25\lambda^2 \\ &= c^2 \end{aligned}$$

$$\Rightarrow \angle C = 90^\circ$$

$$\sin A = \frac{3}{5}, \cos A = \frac{3}{5}, \tan B = \frac{4}{3}, \cot B = \frac{4}{3}$$



3. **Ans. (A,B,C)**

Sol. $\begin{vmatrix} \ln 10 & \ln 100 & \ln 1000 \\ 100 & 200 & 300 \\ a & b & c \end{vmatrix}$

$$= \begin{vmatrix} \ln 10 & 2\ln 10 & 3\ln 10 \\ 100 & 200 & 300 \\ a & b & c \end{vmatrix}$$

$$= (\ln 10)(100) \begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ a & b & c \end{vmatrix} = 0$$

4. **Ans. (A,C,D)**

Sol. In a triangle,

$$\text{we have } \alpha + \beta + \gamma = \pi$$

$$\Rightarrow \tan(\alpha + \beta) = \tan(\pi - \gamma)$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \gamma = \tan \alpha \cdot \tan \beta \cdot \tan \gamma$$

$$\Rightarrow k = 1$$

5. **Ans. (A,B,C,D)**

Sol. Common terms are 16, 37, ... in A.P.

$$a = 16, d = 21$$

$$16 + 21(m-1) \leq 6058$$

$$\Rightarrow m \leq \frac{6058-16}{21} + 1$$

$$\Rightarrow m = 288$$

6. **Ans. (A,B,C)**

Sol. Slope of the required line $= \pm \tan 45^\circ = \pm 1$

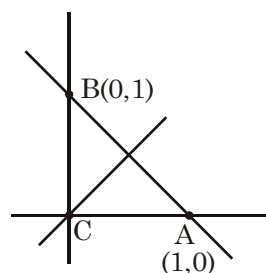
$\Rightarrow$ possible equations

$$x + y = 2 \Rightarrow x + y - 2 = 0$$

$$x - y = 0$$

7. **Ans. (A,B,C)**

Sol.



Point of intersection of $x + y = 1$ and $y = x$

$$\text{is } \left(\frac{1}{2}, \frac{1}{2}\right) \Rightarrow 0 < \alpha < \frac{1}{2}$$

Solution Q.8, Q.9 and Q.10

Sol. $x^2 + 2bx + b^2 = (x + b)^2 \geq 0$

$$\text{Range} = [0, \infty)$$

$$\alpha \times \beta = b^2 \geq 0$$

$$D = 0$$

$$x^2 - 2mx + m^2 - 1 = (x - m)^2 - 1$$

$$\text{Range} = [-1, \infty)$$

$$\alpha \cdot \beta > 0$$

$$D > 0$$

$$x^2 - 6x + 8 = (x - 3)^2 - 1$$

$$\text{Range} = [-1, \infty)$$

$$\alpha \cdot \beta = 8 > 0$$

$$D > 0$$

$$x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\text{Range} = \left[\frac{3}{4}, \infty\right)$$

$$\alpha \cdot \beta = 1 > 0$$

$$D < 0$$

8. **Ans. (C)**

9. **Ans. (B)**

10. **Ans. (D)**

Solution Q.11, Q.12 and Q.13

Sol. (I) $f(x) = \sin^2 x - 4\sin x + 2$

$$f(x) = (\sin x - 2)^2 - 2$$

$$-1 \leq f(x) \leq 7$$

Now $\sin^2 x - 4\sin x + 2 = 1$

$$\Rightarrow \sin^2 x - 4\sin x + 1 = 0$$

$$\Rightarrow \sin x = \frac{4 \pm \sqrt{16-4}}{2}$$

$$\Rightarrow \sin x = 2 \pm \sqrt{3}$$

$$\therefore \sin x = 2 - \sqrt{3}$$

(II) $f(x) = \frac{\cos 3x}{\cos x}$

$$\Rightarrow f(x) = \frac{4\cos^3 x - 3\cos x}{\cos x}$$

$$\Rightarrow f(x) = 4\cos^2 x - 3 \quad (\cos x \neq 0)$$

$$-3 < f(x) \leq 1$$

Now $\frac{\cos 3x}{\cos x} = 1$

$$\Rightarrow 4\cos^3 x - 4\cos x = 0$$

$$\Rightarrow \cos x = 0, \cos x = \pm 1$$

$$\text{but } \cos x \neq 0$$

$$\therefore \cos x = \pm 1$$

(III) $f(x) = \frac{25 + 3\sin x + 4\cos x}{15 - 3\sin x - 4\cos x}$

$$1 \leq f(x) \leq 3$$

Now $\frac{25 + 3\sin x + 4\cos x}{15 - 3\sin x - 4\cos x} = 1$

$$\Rightarrow 10 + 6\sin x + 8\cos x = 0$$

$$\Rightarrow 3\sin x + 4\cos x = -5$$

$$\Rightarrow \frac{3}{5}\sin x + \frac{4}{5}\cos x = -1$$

$$\sin(x + a) = -1, \left(\tan a = \frac{3}{4}, 0 < a < \frac{\pi}{2} \right)$$

$$\therefore \text{ only one solution}$$

(IV) $f(x) = \sin^2 x + 25\operatorname{cosec}^2 x$

$$= (\sin x - \operatorname{cosec} x)^2 + 24\operatorname{cosec}^2 x + 2$$

$$\therefore f(x) \in [26, \infty)$$

Now $\sin^2 x + 25\operatorname{cosec}^2 x = 1$

$$\Rightarrow \sin^2 x + \frac{25}{\sin^2 x} = 1$$

$$\Rightarrow \sin^4 x - \sin^2 x + 25 = 0$$

No solution

11. **Ans. (D)**

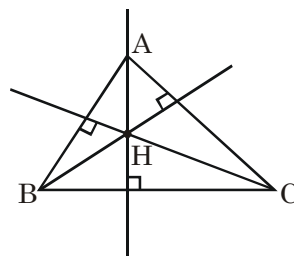
12. **Ans. (C)**

13. **Ans. (D)**

SECTION-II

1. **Ans. 0.75**

Sol.



AC is perpendicular to side BH

HC is perpendicular to side AB

$\Rightarrow$ orthocenter of $\triangle ABH$ is point C

$$\Rightarrow a = 3, b = 4 \Rightarrow \frac{a}{b} = 0.75$$

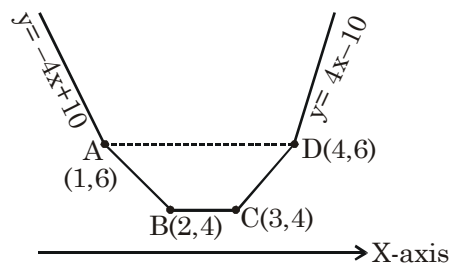
2. **Ans. 0.20**

Sol. Given equation $\Rightarrow x^2 - 3x - 4 = 0$

$$\Rightarrow x = -1, 4 \Rightarrow N = 5$$

3. **Ans. 0.25**

Sol.



Graph of $y = |x-1| + |x-2| + |x-3| + |x-4|$

$$\Rightarrow \lambda = 4 \Rightarrow a = 4 \Rightarrow \frac{1}{a} = 0.25$$

4. Ans. 20.25

Sol. $\frac{a+2b+3c}{3} \geq (a \times 2b \times 3c)^{1/3}$

$$\Rightarrow \frac{64}{27 \times 2 \times 3} \geq abc$$

$$\Rightarrow M = \frac{32}{81}$$

5. Ans. 0.50

Sol. $AD = m_1$

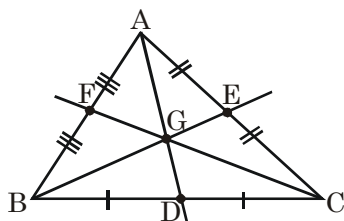
$$BE = m_2$$

$$CF = m_3$$

G is centroid
of $\triangle ABC$

$$BG + GC > BC$$

$$\Rightarrow \frac{2}{3}m_2 + \frac{2}{3}m_3 > a \Rightarrow 2(m_2 + m_3) > 3a$$



Similarly $2(m_3 + m_1) > 3b$

$$2(m_1 + m_2) > 3c$$

$$\Rightarrow 4(m_1 + m_2 + m_3) > 3(a + b + c) \quad \dots(1)$$

In $\triangle ABD$ and $\triangle ADC$

$$c + \frac{a}{2} > m_1 \text{ and } b + \frac{a}{2} > m_1 \Rightarrow a + b + c > 2m_1$$

Similarly $(a + b + c) > 2m_2$

$$(a + b + c) > 2m_3$$

$$\Rightarrow 3(a + b + c) > 2(m_1 + m_2 + m_3) \quad \dots(2)$$

from (1) & (2)

$$2(m_1 + m_2 + m_3) < 3(a + b + c) < 4(m_1 + m_2 + m_3)$$

$$\Rightarrow n = 2$$

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (C)****Sol.** $0^2 = (20)^2 - 2(5)d$

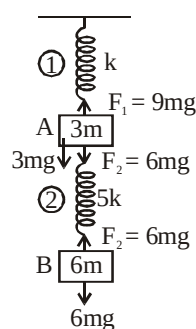
$$d = 40 \text{ m}$$

2. **Ans. (B)****Sol.** (0, 0)

$$x_{\text{cm}} = \frac{m \times 0 + 2m \times \ell/2}{3m} = \frac{\ell}{3}$$

$$y_{\text{cm}} = \frac{2m \times 0 + m \times \ell/2}{3m} = \frac{\ell}{6}$$

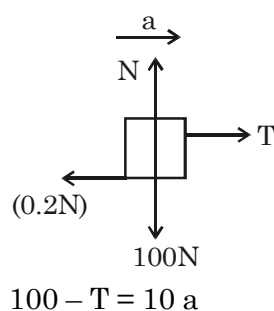
$$\text{COM} = \left(\frac{\ell}{3}, \frac{\ell}{6} \right)$$

3. **Ans. (C)****Sol.** Before Cutting

$$a_a = \frac{9mg - 3mg}{3m} = 2g$$

4. **Ans. (B)**

Sol.

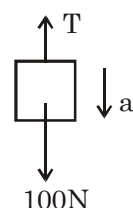


$100 - T = 10a \quad \dots (i)$

$$T - 0.2 \times 100 = 10a \quad \dots (ii)$$

$$N - 100 = 0$$

$$N = 100$$



$$(i) + (ii)$$

$$100 - 20 = 20a$$

$$a = 4 \text{ m/s}^2$$

5. **Ans. (C)****Sol.** One particle will be projected at θ then other will be at $90 - \theta$

$$H_1 = \frac{u^2 \sin^2 \theta}{2g} \quad \dots (i)$$

$$H_2 = \frac{u^2 \sin^2 90 - \theta}{2g} = \frac{u^2 \cos^2 \theta}{2g} \quad \dots (ii)$$

$$H_1 + H_2 = \frac{u^2}{2g}$$

6. **Ans. (C)**

$$\text{Sol. } a_{\text{cm}} = \frac{F_{\text{ext}}}{m_{\text{sys}}} = \frac{F}{3m}$$

7. **Ans. (D)**

$$\text{Sol. } x_{\text{cm}} = \frac{\sum mx}{\sum m}, y_{\text{cm}} = \frac{\sum my}{\sum m}$$

8. **Ans. (B)**

$$\text{Sol. } V_{\text{cm}} = \frac{2 \times 2 + 1(-1)}{2 + 1} = \frac{4 - 1}{3} = 1 \text{ m/s}$$

$$S = Vt$$

$$S = 1 \times 5$$

$$S = 5 \text{ m}$$

9. Ans. (C)

Sol. Here

$$W_{\text{friction}} = -(\mu mg) \times (s_x)$$

$$= -0.5 \times 0.2 \times 10 \times 2$$

$$= -2 \text{ J}$$

$$W_{\text{mg}} = -mgh = -0.2 \times 10 \times 3$$

$$= -6 \text{ J}$$

$$W_F = -(W_{\text{friction}} + W_{\text{mg}})$$

$$= 8 \text{ J}$$

10. Ans. (B)

$$\vec{F} = -\frac{\partial U}{\partial x} \hat{i} - \frac{\partial U}{\partial y} \hat{j} - \frac{\partial U}{\partial z} \hat{k}$$

11. Ans. (A)

Sol. By conservation of linear momentum along the horizontal direction.

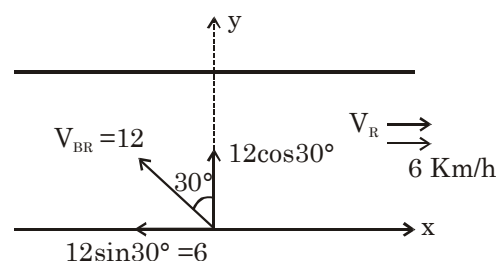
$$(80 + 40) 2.5 = 40 \times v$$

$$v = 7.5 \text{ m/s}$$

$$w = \Delta k = \frac{1}{2} \times 40 \times (7.5)^2 - \frac{1}{2} \times 120 \times (2.5)^2 = 750 \text{ J}$$

12. Ans. (B)

Sol.



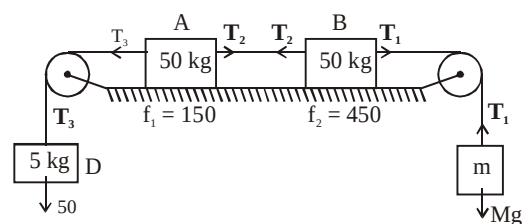
$$\text{Net } V_x = 6 - 6 = 0$$

or boat will appear towards north

13. Ans. (C)

Sol. $T_3 = 50$

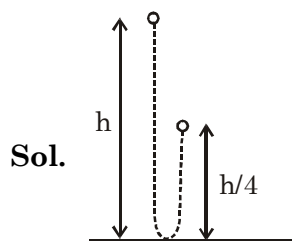
$$T_3 + f_1 = T_2$$



$$T_2 + f_2 = T_1$$

$$T_1 = mg$$

14. Ans. (B)



Sol.

$$\text{Speed just before collision} = \sqrt{2gh} = v_i$$

$$\text{Speed just after collision} = \sqrt{2g \frac{h}{4}} = v_f$$

$$v_f = e v_i$$

$$\frac{\sqrt{2gh}}{2} = e \sqrt{2gh} \Rightarrow e = \frac{1}{2}$$

15. Ans. (B)

$$\text{Sol. } v_1^1 = (1 + e)v_2 - e v_1$$

$$= \left(\frac{3}{2}\right)(-1) - \frac{1}{2}(3) = -\frac{3}{2} - \frac{3}{2} = -3 \text{ m/s}$$

16. Ans. (A)

Sol. No horizontal thrust, velocity of trolley remain constant and horizontal momentum conserved.

17. Ans. (A)

$$\text{Sol. } J - J' = mv$$

$$J' = 2mv$$

$$\Rightarrow v = \frac{J}{3m}$$

18. Ans. (A)

$$\text{Sol. } t \propto m^\alpha b^\beta v_0^\gamma$$

$$M^0 L^0 T^1 = M^\alpha \left[\frac{MLT^{-2}}{LT^{-1}} \right]^\beta \left[LT^{-1} \right]^\gamma$$

$$= M^\alpha [MLT^{-2}]^\beta [LT^{-1}]^{\gamma-n\beta}$$

$$\alpha + \beta = 0 \quad \dots (A)$$

$$\beta + \gamma - n\beta = 0 \quad \dots (B)$$

$$-2\beta - (\gamma - n\beta) = 1 \quad \dots (C)$$

$$-\beta = 1$$

$$\beta = -1$$

$$\text{From (B) } \gamma + n - 1 = 0$$

$$\gamma = 1 - n$$

19. Ans. (B)

20. Ans. (C)

Sol. $F \cdot \Delta t = \Delta P$

SECTION-II

1. Ans. 10.00

Sol. $T = \frac{2u_z}{|a_z|}$

$$T = \frac{2(10 \sin 45)}{(10 \cos 45)}$$

$$T = 2 \text{ sec}$$

or particle will strike plane at 2 sec

Hence speed will be 10 m/s

2. Ans. 8.00

Sol. From work energy theorem $W = \Delta KE$

$$W_{\text{gravity}} + W_{\text{air friction}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\Rightarrow W_{\text{air friction}} = \frac{1}{2}(4)(14)^2 - (4)(10)(10) = -8J$$

$$\Rightarrow \text{Work done against air friction} = 8J$$

3. Ans. 5.00

$$\text{Sol. } a_c = \frac{v^2}{R} = \frac{(30)^2}{300} = 3$$

$$a_t = 4$$

$$a_{\text{total}} = \sqrt{3^2 + 4^2} = 5 \text{ m/s}^2$$

4. Ans. 5.00

Sol. $P = Fv \Rightarrow F = P/v$

$$F = \frac{20}{v}$$

$$\text{or } \frac{20}{v} = \alpha v$$

$$\leftarrow \boxed{} \rightarrow \frac{20}{v} = F$$

$$\Rightarrow \alpha = \frac{20}{v^2} \text{ or } \alpha = \frac{20}{2^2} = 5$$

5. Ans. 15.00

Sol. Under area of $F - t = \Delta P$

so, P will be maximum at $t = 15 \text{ sec}$.

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A)

2. Ans.(D)

3. Ans.(A)

4. Ans.(B)

5. Ans. (C)

6. Ans.(B)

7. Ans. (C)

$$5 \times v = n \times R \times (273) \dots (1)$$

$$P_f \times \frac{V}{3} = n \times R \times (273 + 546) \dots (2)$$

$$P_f = 45$$

8. Ans.(D)

$$\text{Sol. } d = \frac{PM}{RT} \Rightarrow M = \frac{\frac{2 \times 1}{12} \times 600}{2} = 50$$

Relative density with respect to

$$Ne = \frac{\text{Molar mass}}{20}$$

9. Ans (C)

Sol. Volume of cube = $10 \times 10 \times 10 = 1000 \text{ cm}^3$

$$\text{Mass of cube} = 8 \times 1000 = 8000 \text{ gm}$$

$$\text{Moles of Metal} = \frac{6 \times 10^{25}}{6 \times 10^{23}} = 100$$

$$\frac{8000}{\text{atomic weight}} = 100$$

$$\text{atomic weight} = 80$$

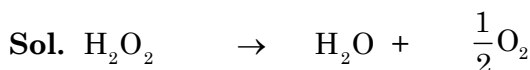
10. Ans (C)

$$\text{Sol. Moles of } T_2O = \frac{5.5}{22}$$

$$\text{Moles of neutrons} = \frac{5.5}{22} \times 12 = 3 \text{ mole}$$

$$\text{No. of neutrons} = 3N_A$$

11. Ans.(B)



$$\text{Used moles} = \frac{1}{2} \text{ mole} \quad \text{moles} = \frac{8}{32} = \frac{1}{4}$$

$$\text{Initial moles of } H_2O_2 = 3 \times 0.5 = 1.5$$

12. Ans. (B)

Sol. Let normally of $KMnO_4$ solution is N

$$\therefore N \times 10 = 50 \times 1 \Rightarrow N = 5$$

molecular equation of MnO_4^- = molecular equation of oxalic acid

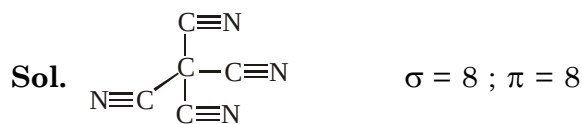
$$5 \times 100 = \frac{W}{E} \times 1000 \Rightarrow \left(\frac{W}{90/2} \right) \times 1000 = W$$

$$= 22.5 \text{ gm Ans.}$$

13. Ans. (B)

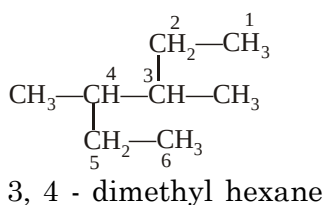
Sol. $M = \frac{4.9 \times 10 \times 0.98}{98}$

14. Ans. (B)



$$\frac{\sigma}{\pi} = 1$$

15. Ans. (D)

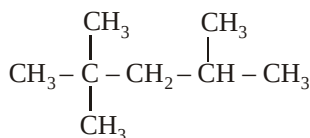


16. Ans. (C)

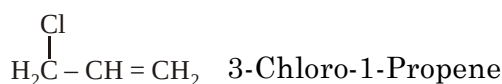
1° & 2° alcohol are same functional group.

17. Ans. (B)

Iso - octane

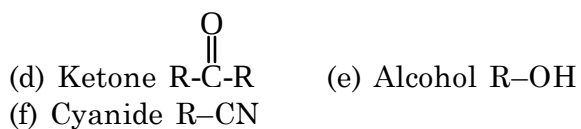
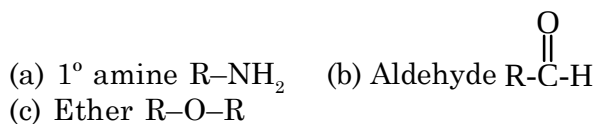


18. Ans. (D)



19. Ans. (C)

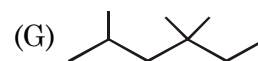
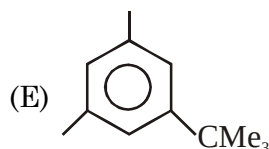
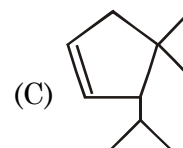
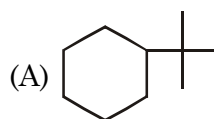
20. Ans. (B)



SECTION-II

1. Ans. (4.00)

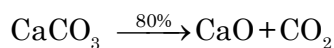
2. Ans. (4.00)



3. Ans (5.00)

Sol. Let mass of impure sample = M gm

$$\text{pure } CaCO_3 = \frac{50}{100} \times M = \frac{M}{2}$$



$$\downarrow$$

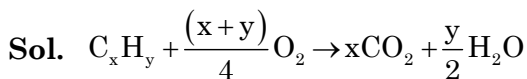
$$\frac{M}{200} \times \frac{80}{100} \text{ moles}$$

$$\frac{M}{200} \times \frac{80}{100} \times 22.4$$

$$= 0.448$$

$$M = 5$$

4. Ans(4.00)



$$x + \frac{y}{4} = 2x$$

$$\frac{y}{4} = x \Rightarrow \frac{y}{x} = 4.00$$

5. Ans. (17.70%) (17.69 to 17.71)

Sol. $\mu_{\text{theoretical}} = q \times d$

$$= \frac{1.6 \times 10^{-19} \times 0.127 \times 10^{-9}}{3.33 \times 10^{-30}} = 6.10 \text{ D}$$

$$\mu_{\text{observed}} = 1.08 \text{ D}$$

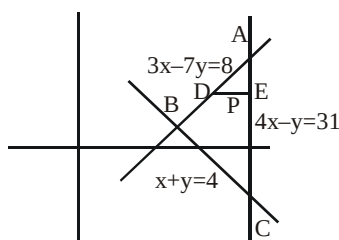
$$\% \text{ Ionic character} = \frac{1.08}{6.10} \times 100 = 17.70\%$$

12. Ans. (A)

$$\begin{aligned} \text{Sol. } D_2 &= \frac{1}{abc} \begin{vmatrix} a & abc & a^2 \\ b & abc & b^2 \\ c & abc & c^2 \end{vmatrix} \\ &= \frac{abc}{abc} \begin{vmatrix} a & 1 & a^2 \\ b & 1 & b^2 \\ c & 1 & c^2 \end{vmatrix} \\ &= \begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ a^2 & b^2 & c^2 \end{vmatrix} \text{ By changing rows to columns} \\ &= - \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix} \text{ By interchanging } R_1 \text{ \& } R_3 \\ &= -D_1 \\ \Rightarrow D_1 + D_2 &= 0 \end{aligned}$$

13. Ans. (A)

Sol.



Equation of DE is $y = 2$
so, point of intersection

$$D\left(\frac{22}{3}, 2\right) \text{ and } E\left(\frac{33}{4}, 2\right)$$

$$\Rightarrow \frac{22}{3} < \lambda < \frac{33}{4}$$

14. Ans. (C)

Sol. For infinite number of solutions

$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & -2 & 1 \\ m & -1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 2(-4 + 1) + 1(2 - m) + 1(-1 + 2m) = 0$$

$$\Rightarrow -6 + 2 - m - 1 + 2m = 0$$

$$\Rightarrow m = 5$$

15. Ans. (B)

$$\text{Sol. } \frac{x_2 - 4}{5} = \frac{y_2 + 13}{1} = -\frac{2(5 \times 4 - 13 + 6)}{5^2 + 1^2} = 1$$

where (x_2, y_2) is image of given point

$$\Rightarrow x_2 - 4 = -5 \Rightarrow x_2 = -1$$

$$\Rightarrow y_2 + 13 = -1 \Rightarrow y_2 = -14$$

16. Ans (D)

$$\text{Sol. } a^{-5}, a^{-4}, a^{-3}, a^{-3}, a^8, a^7$$

$$A.M. \geq G.M.$$

$$\Rightarrow \frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^8 + a^7}{6} \geq [a^{-5} \cdot a^{-4} \cdot a^{-3} \cdot a^{-3} \cdot a^8 \cdot a^7]^{\frac{1}{6}}$$

$$\Rightarrow a^{-5} + a^{-4} + 2a^{-3} + a^8 + a^7 \geq 6$$

{add 6 on both side}

$$\Rightarrow a^{-5} + a^{-4} + 2a^{-3} + a^8 + a^7 + 6 \geq 12$$

So Minimum Value = 12 Ans.

17. Ans. (C)

Sol. Let given points lie on the line

$\ell x + my + n = 0$ where ℓ, m, n are constants
then $\ell(t) + m(2at + at^3) + n = 0$

$$\Rightarrow (am)t^3 + (2am + \ell)t + n = 0$$

t_1, t_2, t_3 are roots of the equation so.

$$t_1 + t_2 + t_3 = 0$$

18. Ans. (C)

$$\text{Sol. } \therefore T_n = \frac{1}{2n \cdot (2n+2) \cdot (2n+4)}$$

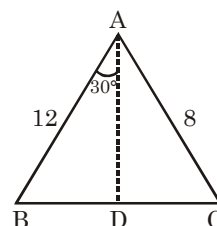
$$T_n = \frac{1}{4} \left[\frac{1}{2n \cdot (2n+2)} - \frac{1}{(2n+2)(2n+4)} \right]$$

$$\therefore S_\infty = \frac{1}{4} \begin{bmatrix} \frac{1}{2.4} & - & \frac{1}{4.6} \\ \frac{1}{4.6} & - & \frac{1}{6.8} \\ \frac{1}{6.8} & - & \frac{1}{8.10} \\ \vdots & & \vdots \\ \infty & & \infty \end{bmatrix}$$

$$\therefore S_\infty = \frac{1}{4} \left[\frac{1}{2.4} \right] = \frac{1}{32}$$

19. Ans. (C)

Sol.



Since $\frac{AB}{AC} = \frac{3}{2} = \frac{BD}{DC}$ hence AD is angle bisector of $\angle BAC$.

$$\ell_{AD} = \frac{2(12)(8)}{(12+8)} \cos 30^\circ = \frac{2(12)(8)}{20} \frac{\sqrt{3}}{2}$$

$$= \frac{24\sqrt{3}}{5}$$

20. Ans. (D)

Sol. We have $||4\sin x| - 1| = 1$
 $\Rightarrow |4\sin x| - 1 = \pm 1 \Rightarrow |4\sin x| = 2 \text{ or } |4\sin x| = 0$

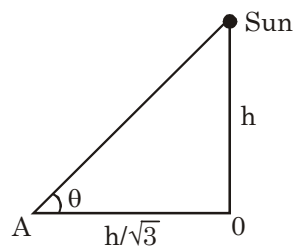
Since $x \in \left(0, \frac{\pi}{2}\right)$ then $|4\sin x| = 4\sin x$

$$\Rightarrow 4\sin x = 2 \Rightarrow \sin x = \frac{2}{4} \Rightarrow x = \frac{\pi}{6}$$

$$\Rightarrow \alpha = \frac{\pi}{6}$$

altitude of sun be θ

$$\tan \theta = \frac{h}{h/\sqrt{3}} = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3} = 2\alpha$$



SECTION-II

1. Ans. 1.50

Sol. $\begin{vmatrix} 1+x & 1 & 1 \\ 3 & 1 & 2 \\ 3 & x & 3 \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} 1+x & 1 & 1 \\ 2+1 & 1 & 2 \\ 3+0 & x & 3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & x & 3 \end{vmatrix} + \begin{vmatrix} x & 1 & 1 \\ 1 & 1 & 2 \\ 0 & x & 3 \end{vmatrix} = 0$$

$$\Rightarrow x(3-2x) - 1(3-x) = 0$$

$$\Rightarrow -2x^2 + 4x - 3 = 0$$

$$\text{Product of roots} = \frac{3}{2} = 1.50$$

2. Ans. 20.00

Sol. Given condition is

$$5a + 5b + 20c = t$$

$$5a + 5b = t - 20c$$

$$\frac{5a}{c-1} + \frac{5b}{c-1} = \frac{t-20c}{c-1}$$

R.H.S. is independent of 'c' if $t = 20$

3. Ans. 1.56 or 1.57

Sol. Given series is an A.G.P.

$$S = 1 + \frac{2}{5} + \frac{3}{5^2} + \frac{4}{5^3} + \dots \infty$$

$$\frac{1}{5}S = \frac{1}{5} + \frac{2}{5^2} + \frac{3}{5^3} + \dots \infty$$

$$S\left(1 - \frac{1}{5}\right) = 1 + \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \dots \infty$$

$$\Rightarrow S \cdot \frac{4}{5} = \frac{1}{1 - \frac{1}{5}} = \frac{5}{4}$$

$$\Rightarrow S = \frac{25}{16} = 1.5625 \text{ Ans.}$$

4. Ans. 3.00

Sol. $\therefore (2)^{\log_2 x^2} - (3)^{\log_3 x} - 6 = 0$

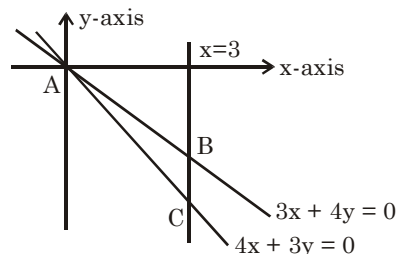
$$\Rightarrow x^2 - x - 6 = 0$$

$$\Rightarrow x = 3 \text{ or } x = -2 \text{ (reject)}$$

$\therefore$ sum of all possible values = 3 Ans.

5. Ans. 0.00

Sol.



Equation of angle bisector of angle A is

$$\frac{3x+4y}{5} = \pm \frac{4x+3y}{5} \Rightarrow x = \pm y$$

$\Rightarrow$ Equation of angle bisector of A is $y = -x$

$\therefore (h, k)$ lies on it so $h + k = 0$

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A, C)**

$$\text{Sol. } mgy = \frac{1}{2}ky^2 \Rightarrow y = \frac{2mg}{k}$$

$$W_{N.C} + W_{mg} + W_{spring} = 0$$

$$W_{NC} = -mg(y_{eq}) + \frac{1}{2}ky_{eq}^2$$

$$y_{eq} = \frac{mg}{k}$$

$$W_{NC} = -\frac{(mg)^2}{2k}$$

2. **Ans. (B,C,D)**

$$\text{Sol. (B) Mass} \propto \frac{1}{(\text{radius})^2}$$

$$\Rightarrow \frac{m}{M_C} = \left(\frac{R}{R/2}\right)^2$$

$$\Rightarrow m_C = \frac{m}{4}$$

Mass of remaining part

$$= m - \frac{m}{4} = \frac{3m}{4}$$

$$(C) y = \frac{m \times 0 - \frac{m}{4} \times \frac{R}{2}}{m - \frac{m}{4}}$$

$$y = -\frac{R}{6}$$

(D) Object is symmetric about x-axis

3. **Ans. (B, C, D)****Sol.** Linear momentum & mechanical energy conserved during collision.

Only conservative forces act during the collision due to which mechanical energy is conserved.

4. **Ans. (A, B, C)****Sol.** (A) Net external force on the given system is zero, so momentum of the system is conserved in ground frame.

$$(B) \vec{P}_{\text{system}} = (m_1 + m_2) \cdot \vec{v}_{\text{com}}$$

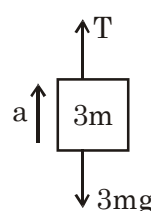
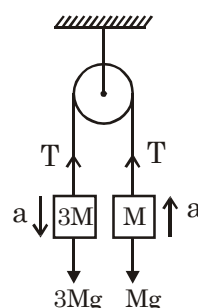
In COM frame $\vec{v}_{\text{com}}$ is equal to zero. $\Rightarrow \vec{P}_{\text{system}}$ in COM frame is always zero.

$$(C) \vec{F}_{\text{Ext}} = 0 \Rightarrow \vec{a}_{\text{Com}} = 0 \Rightarrow \vec{v}_{\text{Com}} \text{ is constant.}$$

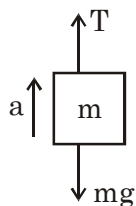
$$KE_{\text{Com}} = \frac{1}{2}(m_1 + m_2) \cdot v_{\text{Com}}^2 \text{ is remains constant.}$$

$$(D) \vec{K} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (\vec{v}_{12})^2$$

KE in COM frame changes into spring potential energy

5. **Ans. (B,C)****Sol.**

$$\Rightarrow 3mg - T = 3ma \dots(i)$$



$$\Rightarrow T - mg = ma \quad \dots(ii)$$

Add eq.(i) + eq.(ii)

$$2mg = 4ma$$

$$\Rightarrow a = g/2$$

From eq.(ii)

$$T = mg + mg/2 = 3mg/2$$

$$\vec{a}_{\text{com}} = \left[\frac{Ma + 3M(-a)}{3M + M} \right] \hat{j} = \frac{-2Ma}{4M} \hat{j} = \frac{-g}{4} \hat{j}$$

$$\vec{F}_{\text{Ext}} = [4M] \vec{a}_{\text{Com}}$$

$$\Rightarrow \vec{F}_{\text{Ext}} \neq 0$$

6. Ans. (B,C)

Sol. Since external force on the system (A + B + plank) in horizontal direction is zero therefore centre of mass of the system remain at rest.

Let displacement of plank is x, then

$$60(4 - x) - 50(4 + x) = 90^\circ x$$

$$40 = 200x$$

$$x = 0.2 \text{ m}$$

7. Ans. (A,C,D)

Sol. For equilibrium

$$F = 0, \frac{-dU}{dx} = 0$$

(A) At point 3

$$F = 0 \text{ and } \frac{dF}{dx} < 0$$

$\Rightarrow$ stable

At point b

$$\frac{dU}{dx} = 0, \frac{d^2U}{dx^2} > 0$$

$\Rightarrow$ stable

(B) At point 2

$F \neq 0$, So there is no equilibrium.

(C) $P = \vec{F} \cdot \vec{V}$

$F = 0$ At point 1, 3 and b

Power delivered by force = zero.

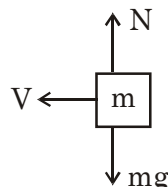
(D) Potential energy is minimum at point b, so KE is maximum.

8. Ans. (D)

Sol. At point C

$$W_g = \Delta KE = \frac{1}{2}mv_c^2 - 0$$

$$v_c^2 = 10Rg \Rightarrow v_c = \sqrt{10Rg}$$



$$\Rightarrow N - mg = \frac{m}{R}[10Rg]$$

$$N = 11mg$$

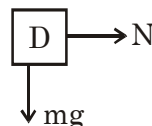
Tangential acceleration (a_t) is zero at point C

9. Ans. (A)

Sol. At point D

$$mg(4R) = \frac{1}{2}mv_D^2 - 0$$

$$v_D = \sqrt{8Rg}$$



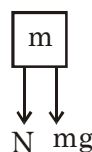
$$\Rightarrow N_D = \frac{mv_D^2}{R} = 8mg$$

$$\Rightarrow mg = ma_t$$

$$a_t = g$$

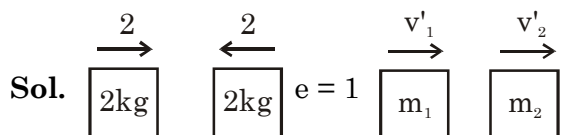
10. Ans. (C)

$$\text{Sol. } mg(3R) = \frac{1}{2}mv_E^2 \Rightarrow v_E = \sqrt{6gR}$$



$$\Rightarrow N + mg = \frac{m(6Rg)}{R} \Rightarrow N = 5mg$$

11. Ans. (A)



$$\vec{v}_{\text{Com}} = \frac{2 \times 2 + 2 \times (-2)}{4} = 0$$

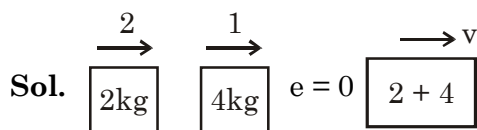
$$v'_2 = 2v_{\text{Com}} - v_2 = 2 \times 2 - (-2) = +2 \text{ m/s}$$

$$KE_{\text{Loss}} = 0 \text{ [For elastic collision]}$$

$$PE_{\text{max}} = [KE]_i - [KE]_{\text{Com}}$$

$$PE_{\text{max}} = 8 \text{ J}$$

12. Ans. (B)



$$2 \times 2 + 4 \times 1 = 6 \times v$$

$$\Rightarrow v = \frac{8}{6} = \frac{4}{3} \text{ m/s}$$

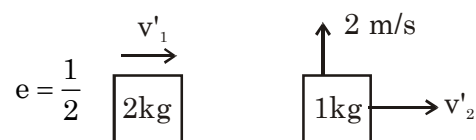
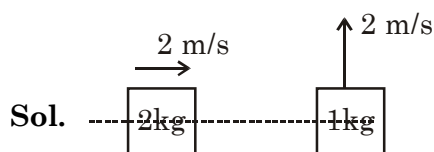
$$KE_i = \frac{1}{2} \times 2(2)^2 + \frac{1}{2} \times 4 \times (1)^2 = 6 \text{ J}$$

$$KE_F = \frac{1}{2} \times 6 \times \left[\frac{4}{3}\right]^2 = 3 \times \frac{16}{9} = \frac{16}{3} \text{ J}$$

$$KE_{\text{Loss}} = KE_i - KE_F = 6 - \frac{16}{3} = \frac{2}{3} \text{ J}$$

$$PE_{\text{max}} = KE_i - KE_{\text{COM}} = \frac{2}{3} \text{ J}$$

13. Ans. (C)



v_{com} along line of impact

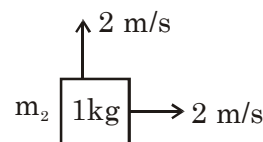
$$v_{\text{com}} = \frac{2 \times 2 + 1 \times 0}{3} = \frac{4}{3} \text{ m/s}$$

$$v'_1 = (1 + e)v_{\text{com}} - ev_1$$

$$= \frac{3}{2} \times \frac{4}{3} - \frac{1}{2} \times 2 = 1 \text{ m/s}$$

$$v'_2 = (1 + e)v_{\text{com}} - ev_2$$

$$= \frac{3}{2} \times \frac{4}{3} = 2 \text{ m/s}$$



$$\sqrt{(2)^2 + (2)^2} = 2\sqrt{2} \text{ m/s}$$

$$KE_i = \frac{1}{2} \times 2(2)^2 + \frac{1}{2} \times 1(2)^2 = 6 \text{ J}$$

$$KE_F = \frac{1}{2} \times 2(1)^2 + \frac{1}{2} \times 1 \times (2\sqrt{2})^2 = 5 \text{ J}$$

$$\Delta KE_{\text{Loss}} = KE_i - KE_F = 6 \text{ J} - 5 \text{ J} = 1 \text{ J}$$

$$PE_{\text{max}} = KE_i - KE_{\text{Com}}$$

$$= \frac{1}{2} \times 2 \times (2)^2 - \frac{1}{2} \times 3 \times \left[\frac{4}{3}\right]^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3} \text{ J}$$

SECTION-II

1. Ans. 3.00

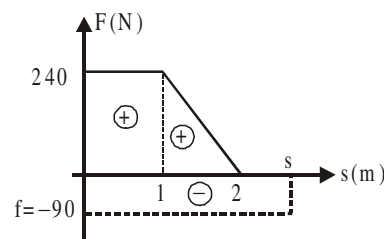
Sol. $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \Rightarrow v_1 + 3v_2 = 4 \dots(i)$

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow v_2 - v_1 = 4 \dots(ii)$$

By solving above equations $v_2 = 2 \text{ m/s}$

Momentum of B = $(1.5)(2) = 3 \text{ kg m/s}$

2. Ans. 4.00



The block will come to rest when total area

will become zero $\frac{1}{2}(20)(1+2) - 90(s) = 0$
 $\Rightarrow s = 4 \text{ m}$

3. Ans. 3.00

Sol. Velocity of B after collision will be

$$V = \frac{3}{2} \frac{2mv_0}{3m} = v_0$$

Tension just after the collision is

$$T = \frac{mv_0^2}{\ell} + mg = 40$$

$$\Rightarrow v_0^2 = 30 \times 0.3$$

$$\Rightarrow v_0 = 3 \text{ m/s}$$

4. Ans. 50.00

Sol. $\frac{1}{2}mu_1^2 + \frac{1}{2}mu_2^2 = \frac{1}{2}kx^2 + \frac{1}{2}(2m)v^2; v = \frac{v_1 + v_2}{2}$

$$\Rightarrow x = \frac{1}{2}m = 50 \text{ cm}$$

5. Ans. 800.00

Sol. By conservation of momentum

$$-200 \times v_1 + 2 \times v_2 = 0$$

$$100 v_1 = v_2 \quad \dots (1)$$

$$v_1 = v_2 / 100$$

$$\left(\frac{v_2}{100} \right) \cdot t = 8$$

$$t = \frac{800}{v_2}$$

$$v_2 \times t = x$$

$$x = v_2 \times \frac{800}{v_2}$$

$$x = 800 \text{ m}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A, B, C, D)

2. Ans.(C,D)



50 ml, 1M 10 ml, 2M

$$1 \times 50 \times \text{n-factor} = 2 \times 10 \times 5$$

$$\text{n-factor} (\text{Sn}^{+2}) = 2$$

So Sn^{+2} is oxidized to Sn^{+4}

3. Ans. (A,B,C,D)

Sol.

$$V = \frac{nRT}{P} : V_{\text{H}_2} = \frac{\left(\frac{1}{2}\right) \times 0.0821 \times (273 + 273)}{1} = 22.4 \ell$$

$$V_{\text{He}} = \frac{\left(\frac{2}{4}\right) \times 0.0821 \times (273 + 273)}{1} = 22.4 \ell$$

$$V_{\text{CH}_4} = \frac{\left(\frac{3}{16}\right) \times 0.0821 \times (273 + 273)}{1} = 8.4 \ell$$

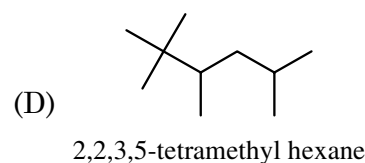
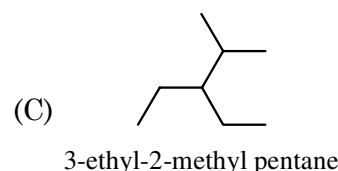
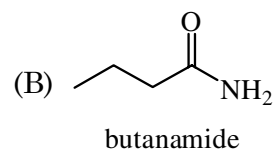
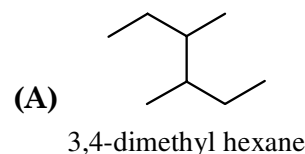
$$V_{\text{SO}_2} = \frac{\left(\frac{4}{64}\right) \times 0.0821 \times (273 + 273)}{1} = 2.8 \ell$$

4. Ans (A, B, D)

5. Ans. (A,D)

6. Ans (A, B, D)

7. Ans. (A, B, C, D)



8. Ans.(C)

9. Ans.(C)

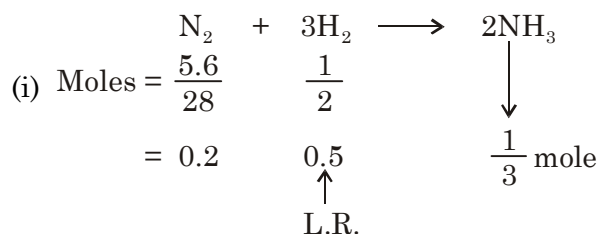
10. Ans.(D)

11. Ans.(B)

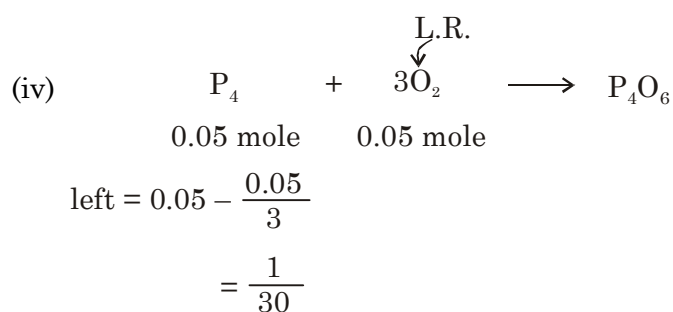
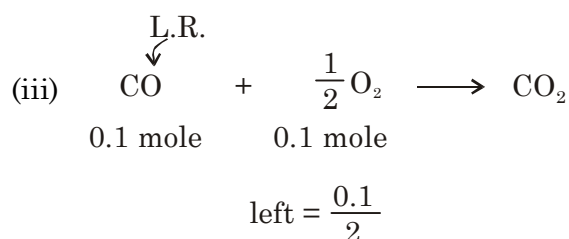
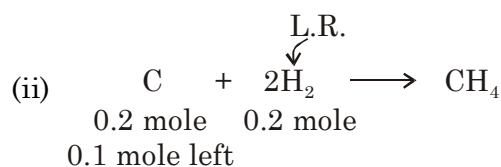
12. Ans.(D)

13. Ans. (A)

Sol.



$$\text{N}_2 \text{ left} = 0.2 - \frac{0.5}{3} = \frac{1}{30}$$



SECTION-II

1. Ans.(2.00)

$$\text{Sol. } P \times V = \frac{4}{M} \times R \times T \rightarrow (1)$$

$$P \times V = \frac{3.2}{M} \times R \times (T+50) \rightarrow (2)$$

$$\frac{(1)}{(2)} \Rightarrow 1 = \frac{4T}{3.2(T+50)}$$

$$T = 200 \text{ K} = 2.00 \times 10^2 \text{ K}$$

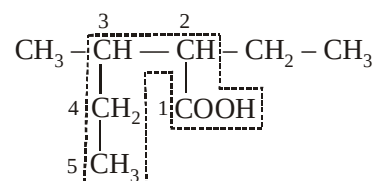
2. Ans (7.00)

number of Ring = 5

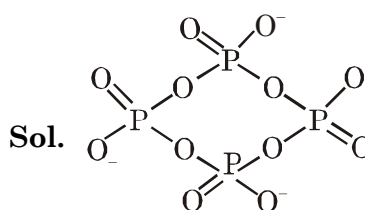
number of II bond = 2

$$\therefore \text{DBE} = 7$$

3. Ans. (5.00)



4. Ans.(4.00)



5. Ans. (13.15)

$$\text{Sol. } x = 4; y = 21; z = 1.30$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (B,C,D)**

Sol. System has unique solution

$$\Rightarrow \begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & \lambda \\ 1 & \lambda & 1 \end{vmatrix} \neq 0 \Rightarrow \lambda \neq -\frac{1}{2}, 1$$

2. **Ans. (A,D)**

Sol. $\frac{a}{3} = \frac{b}{4} = \frac{c}{5} = \lambda$

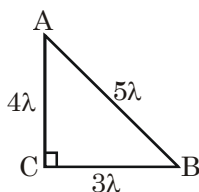
$$a = 3\lambda, b = 4\lambda, c = 5\lambda,$$

$$\lambda > 0$$

$$\begin{aligned} \text{Hence } a^2 + b^2 &= 9\lambda^2 + 16\lambda^2 \\ &= 25\lambda^2 \\ &= c^2 \end{aligned}$$

$$\Rightarrow \angle C = 90^\circ$$

$$\sin A = \frac{3}{5}, \cos A = \frac{3}{5}, \tan B = \frac{4}{3}, \cot B = \frac{4}{3}$$



3. **Ans. (A,B,C)**

Sol. $\begin{vmatrix} \ln 10 & \ln 100 & \ln 1000 \\ 100 & 200 & 300 \\ a & b & c \end{vmatrix}$

$$= \begin{vmatrix} \ln 10 & 2\ln 10 & 3\ln 10 \\ 100 & 200 & 300 \\ a & b & c \end{vmatrix}$$

$$= (\ln 10)(100) \begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ a & b & c \end{vmatrix} = 0$$

4. **Ans. (A,C,D)**

Sol. In a triangle,

$$\text{we have } \alpha + \beta + \gamma = \pi$$

$$\Rightarrow \tan(\alpha + \beta) = \tan(\pi - \gamma)$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \gamma = \tan \alpha \cdot \tan \beta \cdot \tan \gamma$$

$$\Rightarrow k = 1$$

5. **Ans. (A,B,C,D)**

Sol. Common terms are 16, 37, ... in A.P.

$$a = 16, d = 21$$

$$16 + 21(m - 1) \leq 6058$$

$$\Rightarrow m \leq \frac{6058 - 16}{21} + 1$$

$$\Rightarrow m = 288$$

6. **Ans. (A,B,C)**

Sol. Slope of the required line $= \pm \tan 45^\circ = \pm 1$

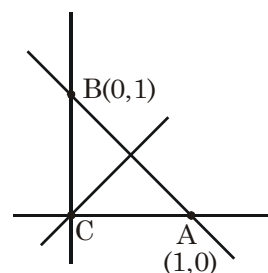
$\Rightarrow$ possible equations

$$x + y = 2 \Rightarrow x + y - 2 = 0$$

$$x - y = 0$$

7. **Ans. (A,B,C)**

Sol.



Point of intersection of $x + y = 1$ and $y = x$

$$\text{is } \left(\frac{1}{2}, \frac{1}{2}\right) \Rightarrow 0 < \alpha < \frac{1}{2}$$

Solution Q.8, Q.9 and Q.10

Sol. $x^2 + 2bx + b^2 = (x + b)^2 \geq 0$

$$\text{Range} = [0, \infty)$$

$$\alpha \times \beta = b^2 \geq 0$$

$$D = 0$$

$$x^2 - 2mx + m^2 - 1 = (x - m)^2 - 1$$

$$\text{Range} = [-1, \infty)$$

$$\alpha \cdot \beta > 0$$

$$D > 0$$

$$x^2 - 6x + 8 = (x - 3)^2 - 1$$

$$\text{Range} = [-1, \infty)$$

$$\alpha \cdot \beta = 8 > 0$$

$$D > 0$$

$$x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\text{Range} = \left[\frac{3}{4}, \infty\right)$$

$$\alpha \cdot \beta = 1 > 0$$

$$D < 0$$

8. **Ans. (C)**

9. **Ans. (B)**

10. **Ans. (D)**

Solution Q.11, Q.12 and Q.13

Sol. (I) $f(x) = \sin^2 x - 4\sin x + 2$

$$f(x) = (\sin x - 2)^2 - 2$$

$$-1 \leq f(x) \leq 7$$

Now $\sin^2 x - 4\sin x + 2 = 1$

$$\Rightarrow \sin^2 x - 4\sin x + 1 = 0$$

$$\Rightarrow \sin x = \frac{4 \pm \sqrt{16-4}}{2}$$

$$\Rightarrow \sin x = 2 \pm \sqrt{3}$$

$$\therefore \sin x = 2 - \sqrt{3}$$

(II) $f(x) = \frac{\cos 3x}{\cos x}$

$$\Rightarrow f(x) = \frac{4\cos^3 x - 3\cos x}{\cos x}$$

$$\Rightarrow f(x) = 4\cos^2 x - 3 \quad (\cos x \neq 0)$$

$$-3 < f(x) \leq 1$$

Now $\frac{\cos 3x}{\cos x} = 1$

$$\Rightarrow 4\cos^3 x - 4\cos x = 0$$

$$\Rightarrow \cos x = 0, \cos x = \pm 1$$

$$\text{but } \cos x \neq 0$$

$$\therefore \cos x = \pm 1$$

(III) $f(x) = \frac{25 + 3\sin x + 4\cos x}{15 - 3\sin x - 4\cos x}$

$$1 \leq f(x) \leq 3$$

Now $\frac{25 + 3\sin x + 4\cos x}{15 - 3\sin x - 4\cos x} = 1$

$$\Rightarrow 10 + 6\sin x + 8\cos x = 0$$

$$\Rightarrow 3\sin x + 4\cos x = -5$$

$$\Rightarrow \frac{3}{5}\sin x + \frac{4}{5}\cos x = -1$$

$$\sin(x + a) = -1, \left(\tan a = \frac{3}{4}, 0 < a < \frac{\pi}{2} \right)$$

$$\therefore \text{ only one solution}$$

(IV) $f(x) = \sin^2 x + 25\operatorname{cosec}^2 x$

$$= (\sin x - \operatorname{cosec} x)^2 + 24\operatorname{cosec}^2 x + 2$$

$$\therefore f(x) \in [26, \infty)$$

Now $\sin^2 x + 25\operatorname{cosec}^2 x = 1$

$$\Rightarrow \sin^2 x + \frac{25}{\sin^2 x} = 1$$

$$\Rightarrow \sin^4 x - \sin^2 x + 25 = 0$$

No solution

11. **Ans. (D)**

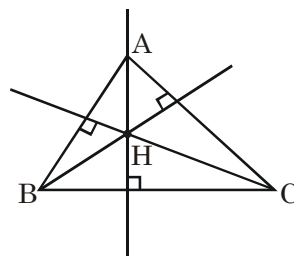
12. **Ans. (C)**

13. **Ans. (D)**

SECTION-II

1. **Ans. 0.75**

Sol.



AC is perpendicular to side BH

HC is perpendicular to side AB

$\Rightarrow$ orthocenter of $\triangle ABH$ is point C

$$\Rightarrow a = 3, b = 4 \Rightarrow \frac{a}{b} = 0.75$$

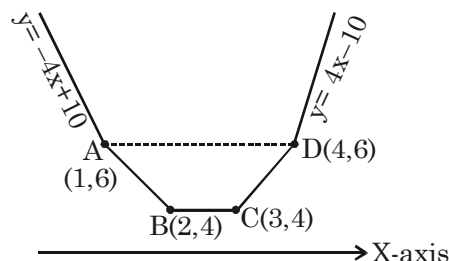
2. **Ans. 0.20**

Sol. Given equation $\Rightarrow x^2 - 3x - 4 = 0$

$$\Rightarrow x = -1, 4 \Rightarrow N = 5$$

3. **Ans. 0.25**

Sol.



Graph of $y = |x-1| + |x-2| + |x-3| + |x-4|$

$$\Rightarrow \lambda = 4 \Rightarrow a = 4 \Rightarrow \frac{1}{a} = 0.25$$

4. Ans. 20.25

Sol. $\frac{a+2b+3c}{3} \geq (a \times 2b \times 3c)^{1/3}$

$$\Rightarrow \frac{64}{27 \times 2 \times 3} \geq abc$$

$$\Rightarrow M = \frac{32}{81}$$

5. Ans. 0.50

Sol. $AD = m_1$

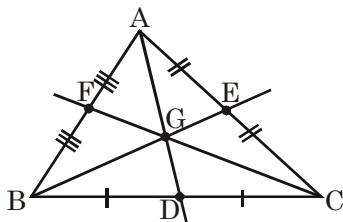
$$BE = m_2$$

$$CF = m_3$$

G is centroid
of $\triangle ABC$

$$BG + GC > BC$$

$$\Rightarrow \frac{2}{3}m_2 + \frac{2}{3}m_3 > a \Rightarrow 2(m_2 + m_3) > 3a$$



Similarly $2(m_3 + m_1) > 3b$

$$2(m_1 + m_2) > 3c$$

$$\Rightarrow 4(m_1 + m_2 + m_3) > 3(a + b + c) \quad \dots(1)$$

In $\triangle ABD$ and $\triangle ADC$

$$c + \frac{a}{2} > m_1 \text{ and } b + \frac{a}{2} > m_1 \Rightarrow a + b + c > 2m_1$$

Similarly $(a + b + c) > 2m_2$

$$(a + b + c) > 2m_3$$

$$\Rightarrow 3(a + b + c) > 2(m_1 + m_2 + m_3) \quad \dots(2)$$

from (1) & (2)

$$2(m_1 + m_2 + m_3) < 3(a + b + c) < 4(m_1 + m_2 + m_3)$$

$$\Rightarrow n = 2$$

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (C)****Sol.** $0^2 = (20)^2 - 2(5)d$

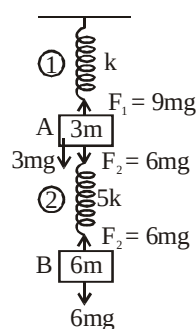
$$d = 40 \text{ m}$$

2. **Ans. (B)****Sol.** (0, 0)

$$x_{\text{cm}} = \frac{m \times 0 + 2m \times \ell/2}{3m} = \frac{\ell}{3}$$

$$y_{\text{cm}} = \frac{2m \times 0 + m \times \ell/2}{3m} = \frac{\ell}{6}$$

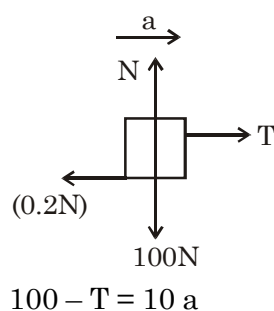
$$\text{COM} = \left(\frac{\ell}{3}, \frac{\ell}{6} \right)$$

3. **Ans. (C)****Sol.** Before Cutting

$$a_a = \frac{9mg - 3mg}{3m} = 2g$$

4. **Ans. (B)**

Sol.

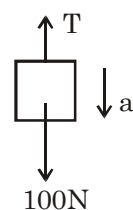


$$100 - T = 10a \quad \dots (i)$$

$$T - 0.2 \times 100 = 10a \quad \dots (ii)$$

$$N - 100 = 0$$

$$N = 100$$



$$(i) + (ii)$$

$$100 - 20 = 20a$$

$$a = 4 \text{ m/s}^2$$

5. **Ans. (C)****Sol.** One particle will be projected at θ then other will be at $90 - \theta$

$$H_1 = \frac{u^2 \sin^2 \theta}{2g} \quad \dots (i)$$

$$H_2 = \frac{u^2 \sin^2 90 - \theta}{2g} = \frac{u^2 \cos^2 \theta}{2g} \quad \dots (ii)$$

$$H_1 + H_2 = \frac{u^2}{2g}$$

6. **Ans. (C)**

$$\text{Sol. } a_{\text{cm}} = \frac{F_{\text{ext}}}{m_{\text{sys}}} = \frac{F}{3m}$$

7. **Ans. (D)**

$$\text{Sol. } x_{\text{cm}} = \frac{\sum mx}{\sum m}, y_{\text{cm}} = \frac{\sum my}{\sum m}$$

8. **Ans. (B)**

$$\text{Sol. } V_{\text{cm}} = \frac{2 \times 2 + 1(-1)}{2 + 1} = \frac{4 - 1}{3} = 1 \text{ m/s}$$

$$S = Vt$$

$$S = 1 \times 5$$

$$S = 5 \text{ m}$$

9. Ans. (C)

Sol. Here

$$\begin{aligned} W_{\text{friction}} &= -(\mu mg) \times (s_x) \\ &= -0.5 \times 0.2 \times 10 \times 2 \\ &= -2 \text{ J} \\ W_{\text{mg}} &= -mgh = -0.2 \times 10 \times 3 \\ &= -6 \text{ J} \\ W_F &= -(W_{\text{friction}} + W_{\text{mg}}) \\ &= 8 \text{ J} \end{aligned}$$

10. Ans. (B)

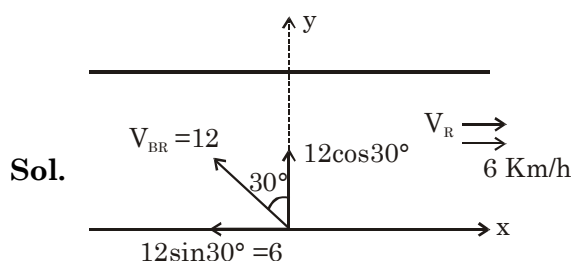
Sol. $\vec{F} = -\frac{\partial U}{\partial x} \hat{i} - \frac{\partial U}{\partial y} \hat{j} - \frac{\partial U}{\partial z} \hat{k}$

11. Ans. (A)

Sol. By conservation of linear momentum along the horizontal direction.

$$\begin{aligned} (80 + 40) 2.5 &= 40 \times v \\ v &= 7.5 \text{ m/s} \\ w = \Delta k &= \frac{1}{2} \times 40 \times (7.5)^2 \\ &\quad - \frac{1}{2} \times 120 \times (2.5)^2 = 750 \text{ J} \end{aligned}$$

12. Ans. (B)



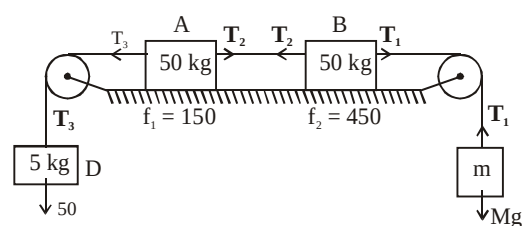
$$\text{Net } V_x = 6 - 6 = 0$$

or boat will appear towards north

13. Ans. (C)

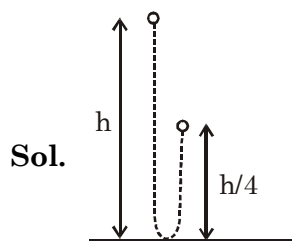
Sol. $T_3 = 50$

$$T_3 + f_1 = T_2$$



$$\begin{aligned} T_2 + f_2 &= T_1 \\ T_1 &= mg \end{aligned}$$

14. Ans. (B)



$$\text{Speed just before collision} = \sqrt{2gh} = v_i$$

$$\text{Speed just after collision} = \sqrt{2g \frac{h}{4}} = v_f$$

$$v_f = e v_i$$

$$\frac{\sqrt{2gh}}{2} = e \sqrt{2gh} \Rightarrow e = \frac{1}{2}$$

15. Ans. (B)

Sol. $v_1^1 = (1 + e)v_2 - e v_1$

$$= \left(\frac{3}{2}\right)(-1) - \frac{1}{2}(3) = -\frac{3}{2} - \frac{3}{2} = -3 \text{ m/s}$$

16. Ans. (A)

Sol. No horizontal thrust, velocity of trolley remain constant and horizontal momentum conserved.

17. Ans. (A)

Sol. $J - J' = mv$

$$J' = 2mv$$

$$\Rightarrow v = \frac{J}{3m}$$

18. Ans. (A)

Sol. $t \propto m^\alpha b^\beta v_0^\gamma$

$$M^0 L^0 T^1 = M^\alpha \left[\frac{MLT^{-2}}{LT^{-1}} \right]^\beta \left[LT^{-1} \right]^\gamma$$

$$= M^\alpha [MLT^{-2}]^\beta [LT^{-1}]^{\gamma-n\beta}$$

$$\alpha + \beta = 0 \quad \dots (A)$$

$$\beta + \gamma - n\beta = 0 \quad \dots (B)$$

$$-2\beta - (\gamma - n\beta) = 1 \quad \dots (C)$$

$$-\beta = 1$$

$$\beta = -1$$

$$\text{From (B)} \quad \gamma + n - 1 = 0$$

$$\gamma = 1 - n$$

19. Ans. (B)

20. Ans. (C)

Sol. $F \cdot \Delta t = \Delta P$

SECTION-II

1. Ans. 10.00

Sol. $T = \frac{2u_z}{|a_z|}$

$$T = \frac{2(10 \sin 45)}{(10 \cos 45)}$$

$$T = 2 \text{ sec}$$

or particle will strike plane at 2 sec

Hence speed will be 10 m/s

2. Ans. 8.00

Sol. From work energy theorem $W = \Delta KE$

$$W_{\text{gravity}} + W_{\text{air friction}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\Rightarrow W_{\text{air friction}} = \frac{1}{2}(4)(14)^2 - (4)(10)(10) = -8J$$

$$\Rightarrow \text{Work done against air friction} = 8J$$

3. Ans. 5.00

$$\text{Sol. } a_c = \frac{v^2}{R} = \frac{(30)^2}{300} = 3$$

$$a_t = 4$$

$$a_{\text{total}} = \sqrt{3^2 + 4^2} = 5 \text{ m/s}^2$$

4. Ans. 5.00

Sol. $P = Fv \Rightarrow F = P/v$

$$F = \frac{20}{v}$$

$$\text{or } \frac{20}{v} = \alpha v$$

$$\leftarrow \boxed{} \rightarrow \frac{20}{v} = F$$

$$\Rightarrow \alpha = \frac{20}{v^2} \text{ or } \alpha = \frac{20}{2^2} = 5$$

5. Ans. 15.00

Sol. Under area of $F - t = \Delta P$

so, P will be maximum at $t = 15 \text{ sec}$.

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A)

2. Ans.(D)

3. Ans.(A)

4. Ans.(B)

5. Ans. (C)

6. Ans.(B)

7. Ans. (C)

$$5 \times v = n \times R \times (273) \dots (1)$$

$$P_f \times \frac{V}{3} = n \times R \times (273 + 546) \dots (2)$$

$$P_f = 45$$

8. Ans.(D)

$$\text{Sol. } d = \frac{PM}{RT} \Rightarrow M = \frac{\frac{2 \times 1}{12} \times 600}{2} = 50$$

Relative density with respect to

$$Ne = \frac{\text{Molar mass}}{20}$$

9. Ans (C)

Sol. Volume of cube = $10 \times 10 \times 10 = 1000 \text{ cm}^3$

$$\text{Mass of cube} = 8 \times 1000 = 8000 \text{ gm}$$

$$\text{Moles of Metal} = \frac{6 \times 10^{25}}{6 \times 10^{23}} = 100$$

$$\frac{8000}{\text{atomic weight}} = 100$$

$$\text{atomic weight} = 80$$

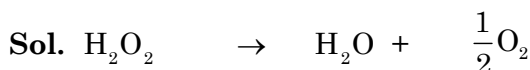
10. Ans (C)

$$\text{Sol. Moles of } T_2O = \frac{5.5}{22}$$

$$\text{Moles of neutrons} = \frac{5.5}{22} \times 12 = 3 \text{ mole}$$

$$\text{No. of neutrons} = 3N_A$$

11. Ans.(B)



$$\text{Used moles} = \frac{1}{2} \text{ mole} \quad \text{moles} = \frac{8}{32} = \frac{1}{4}$$

$$\text{Initial moles of } H_2O_2 = 3 \times 0.5 = 1.5$$

12. Ans. (B)

Sol. Let normally of $KMnO_4$ solution is N

$$\therefore N \times 10 = 50 \times 1 \Rightarrow N = 5$$

molecular equation of MnO_4^- = molecular equation of oxalic acid

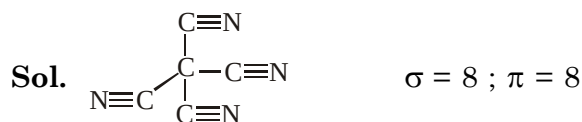
$$5 \times 100 = \frac{W}{E} \times 1000 \Rightarrow \left(\frac{W}{90/2} \right) \times 1000 W$$

$$= 22.5 \text{ gm Ans.}$$

13. Ans.(B)

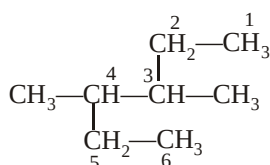
Sol. $M = \frac{4.9 \times 10 \times 0.98}{98}$

14. Ans. (B)



$$\frac{\sigma}{\pi} = 1$$

15. Ans. (D)



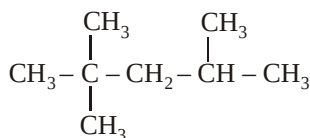
3, 4 - dimethyl hexane

16. Ans. (C)

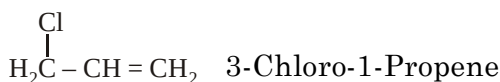
1° & 2° alcohol are same functional group.

17. Ans. (B)

Iso - octane



18. Ans. (D)



19. Ans. (C)

20. Ans. (B)

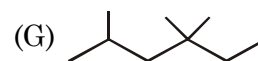
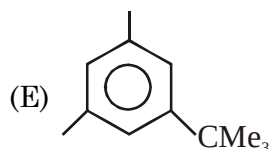
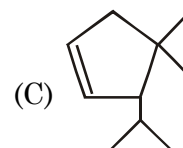
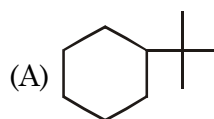
(a) 1° amine $R-NH_2$ (b) Aldehyde $R-\overset{\overset{O}{||}}{C}-H$
(c) Ether $R-O-R$

(d) Ketone $R-\overset{\overset{O}{||}}{C}-R$ (e) Alcohol $R-OH$
(f) Cyanide $R-CN$

SECTION-II

1. Ans. (4.00)

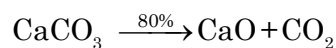
2. Ans. (4.00)



3. Ans (5.00)

Sol. Let mass of impure sample = M gm

$$\text{pure } CaCO_3 = \frac{50}{100} \times M = \frac{M}{2}$$



$$\downarrow$$

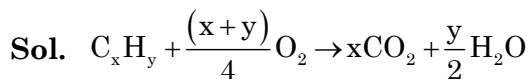
$$\frac{M}{200} \times \frac{80}{100} \text{ moles}$$

$$\frac{M}{200} \times \frac{80}{100} \times 22.4$$

$$= 0.448$$

$$M = 5$$

4. Ans(4.00)



$$x + \frac{y}{4} = 2x$$

$$\frac{y}{4} = x \Rightarrow \frac{y}{x} = 4.00$$

5. Ans. (17.70%) (17.69 to 17.71)

Sol. $\mu_{\text{theoretical}} = q \times d$

$$= \frac{1.6 \times 10^{-19} \times 0.127 \times 10^{-9}}{3.33 \times 10^{-30}} = 6.10 \text{ D}$$

$$\mu_{\text{observed}} = 1.08 \text{ D}$$

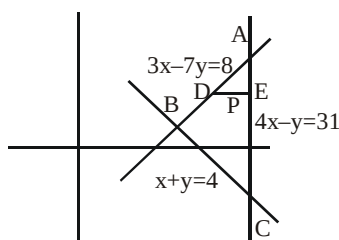
$$\% \text{ Ionic character} = \frac{1.08}{6.10} \times 100 = 17.70\%$$

12. Ans. (A)

$$\begin{aligned} \text{Sol. } D_2 &= \frac{1}{abc} \begin{vmatrix} a & abc & a^2 \\ b & abc & b^2 \\ c & abc & c^2 \end{vmatrix} \\ &= \frac{abc}{abc} \begin{vmatrix} a & 1 & a^2 \\ b & 1 & b^2 \\ c & 1 & c^2 \end{vmatrix} \\ &= \begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ a^2 & b^2 & c^2 \end{vmatrix} \text{ By changing rows to columns} \\ &= - \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix} \text{ By interchanging } R_1 \text{ \& } R_3 \\ &= -D_1 \\ \Rightarrow D_1 + D_2 &= 0 \end{aligned}$$

13. Ans. (A)

Sol.



Equation of DE is $y = 2$
so, point of intersection

$$D\left(\frac{22}{3}, 2\right) \text{ and } E\left(\frac{33}{4}, 2\right)$$

$$\Rightarrow \frac{22}{3} < \lambda < \frac{33}{4}$$

14. Ans. (C)

Sol. For infinite number of solutions

$$\begin{vmatrix} 2 & -1 & 1 \\ 1 & -2 & 1 \\ m & -1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 2(-4 + 1) + 1(2 - m) + 1(-1 + 2m) = 0$$

$$\Rightarrow -6 + 2 - m - 1 + 2m = 0$$

$$\Rightarrow m = 5$$

15. Ans. (B)

$$\text{Sol. } \frac{x_2 - 4}{5} = \frac{y_2 + 13}{1} = -\frac{2(5 \times 4 - 13 + 6)}{5^2 + 1^2} = 1$$

where (x_2, y_2) is image of given point

$$\Rightarrow x_2 - 4 = -5 \Rightarrow x_2 = -1$$

$$\Rightarrow y_2 + 13 = -1 \Rightarrow y_2 = -14$$

16. Ans (D)

$$\text{Sol. } a^{-5}, a^{-4}, a^{-3}, a^{-3}, a^8, a^7$$

$$A.M. \geq G.M.$$

$$\Rightarrow \frac{a^{-5} + a^{-4} + a^{-3} + a^{-3} + a^8 + a^7}{6} \geq [a^{-5} \cdot a^{-4} \cdot a^{-3} \cdot a^{-3} \cdot a^8 \cdot a^7]^{\frac{1}{6}}$$

$$\Rightarrow a^{-5} + a^{-4} + 2a^{-3} + a^8 + a^7 \geq 6$$

{add 6 on both side}

$$\Rightarrow a^{-5} + a^{-4} + 2a^{-3} + a^8 + a^7 + 6 \geq 12$$

So Minimum Value = 12 Ans.

17. Ans. (C)

Sol. Let given points lie on the line

$\ell x + my + n = 0$ where ℓ, m, n are constants
then $\ell(t) + m(2at + at^3) + n = 0$

$$\Rightarrow (am)t^3 + (2am + \ell)t + n = 0$$

t_1, t_2, t_3 are roots of the equation so.

$$t_1 + t_2 + t_3 = 0$$

18. Ans. (C)

$$\text{Sol. } \therefore T_n = \frac{1}{2n \cdot (2n+2) \cdot (2n+4)}$$

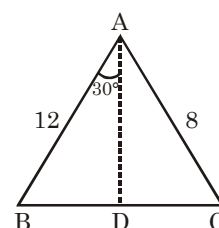
$$T_n = \frac{1}{4} \left[\frac{1}{2n \cdot (2n+2)} - \frac{1}{(2n+2)(2n+4)} \right]$$

$$\therefore S_\infty = \frac{1}{4} \begin{bmatrix} \frac{1}{2.4} & - & \frac{1}{4.6} \\ \frac{1}{4.6} & - & \frac{1}{6.8} \\ \frac{1}{6.8} & - & \frac{1}{8.10} \\ \vdots & & \vdots \\ \infty & & \infty \end{bmatrix}$$

$$\therefore S_\infty = \frac{1}{4} \left[\frac{1}{2.4} \right] = \frac{1}{32}$$

19. Ans. (C)

Sol.



Since $\frac{AB}{AC} = \frac{3}{2} = \frac{BD}{DC}$ hence AD is angle bisector of $\angle BAC$.

$$\ell_{AD} = \frac{2(12)(8)}{(12+8)} \cos 30^\circ = \frac{2(12)(8)}{20} \frac{\sqrt{3}}{2}$$

$$= \frac{24\sqrt{3}}{5}$$

20. Ans. (D)

Sol. We have $||4\sin x| - 1| = 1$
 $\Rightarrow |4\sin x| - 1 = \pm 1 \Rightarrow |4\sin x| = 2 \text{ or } |4\sin x| = 0$

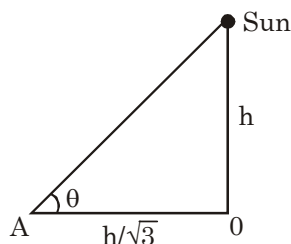
Since $x \in \left(0, \frac{\pi}{2}\right)$ then $|4\sin x| = 4\sin x$

$$\Rightarrow 4\sin x = 2 \Rightarrow \sin x = \frac{2}{4} \Rightarrow x = \frac{\pi}{6}$$

$$\Rightarrow \alpha = \frac{\pi}{6}$$

altitude of sun be θ

$$\tan \theta = \frac{h}{h/\sqrt{3}} = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3} = 2\alpha$$



SECTION-II

1. Ans. 1.50

Sol. $\begin{vmatrix} 1+x & 1 & 1 \\ 3 & 1 & 2 \\ 3 & x & 3 \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} 1+x & 1 & 1 \\ 2+1 & 1 & 2 \\ 3+0 & x & 3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & x & 3 \end{vmatrix} + \begin{vmatrix} x & 1 & 1 \\ 1 & 1 & 2 \\ 0 & x & 3 \end{vmatrix} = 0$$

$$\Rightarrow x(3-2x) - 1(3-x) = 0$$

$$\Rightarrow -2x^2 + 4x - 3 = 0$$

$$\text{Product of roots} = \frac{3}{2} = 1.50$$

2. Ans. 20.00

Sol. Given condition is

$$5a + 5b + 20c = t$$

$$5a + 5b = t - 20c$$

$$\frac{5a}{c-1} + \frac{5b}{c-1} = \frac{t-20c}{c-1}$$

R.H.S. is independent of 'c' if $t = 20$

3. Ans. 1.56 or 1.57

Sol. Given series is an A.G.P.

$$S = 1 + \frac{2}{5} + \frac{3}{5^2} + \frac{4}{5^3} + \dots \infty$$

$$\frac{1}{5}S = \frac{1}{5} + \frac{2}{5^2} + \frac{3}{5^3} + \dots \infty$$

$$S\left(1 - \frac{1}{5}\right) = 1 + \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \dots \infty$$

$$\Rightarrow S \cdot \frac{4}{5} = \frac{1}{1 - \frac{1}{5}} = \frac{5}{4}$$

$$\Rightarrow S = \frac{25}{16} = 1.5625 \text{ Ans.}$$

4. Ans. 3.00

Sol. $\therefore (2)^{\log_2 x^2} - (3)^{\log_3 x} - 6 = 0$

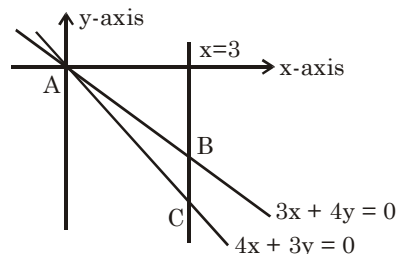
$$\Rightarrow x^2 - x - 6 = 0$$

$$\Rightarrow x = 3 \text{ or } x = -2 \text{ (reject)}$$

$\therefore$ sum of all possible values = 3 Ans.

5. Ans. 0.00

Sol.



Equation of angle bisector of angle A is

$$\frac{3x+4y}{5} = \pm \frac{4x+3y}{5} \Rightarrow x = \pm y$$

$\Rightarrow$ Equation of angle bisector of A is $y = -x$

$\therefore (h, k)$ lies on it so $h + k = 0$